

READING ATTENTION AT A DEPTH THAT DOES NOT MATTER: TRANSPORT-INFORMED TOKEN ALLOCATION FOR VISION–LANGUAGE MODELS

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Paper under double-blind review

ABSTRACT

Vision–language models on high-resolution imagery must decide where to spend a fixed visual-token budget, a decision that determines whether fine-grained evidence survives encoding at all. The prevailing remedy is attention-guided cropping: read the model’s own attention map, crop where it peaks, re-encode. Every such method must choose a depth at which to read that map, and published methods hand-pick a single layer or average a fixed block. We show this choice dominates every other design decision — single-layer read-out collapses to the four-option chance rate — and that the depth these methods read is not the depth at which the image reaches the answer. Layer-wise masking of text-to-image attention localises that transport to the first two thirds of the stack: blocking every layer from layer 16 of 28 onward flips no answers, while blocking layers 0–16 reproduces the full effect. The boundary transfers across architectures as a fixed fraction of depth. Two mechanisms follow. A single closed-form ridge solve over the 28-layer attention profile yields a signed depth filter that subtracts the late layers the conventional read-out adds, improving crop placement by 7.9 and 8.9 points over the strongest published rule at matched compute. Because visual tokens are causally inert past the boundary, 90% of them can be discarded there at negligible cost under any ranking criterion, and the freed compute reallocated to encoding resolution, where it recovers the model’s resolution headroom at a rate of 1.01 ($r = 0.966$) across four checkpoints and three benchmarks. Against a model running at its *native* dynamic resolution — $5.5\times$ our budget — our methods are statistically indistinguishable while using 18% of the compute, whereas published allocation rules fall 12–14 points short of it.

1 INTRODUCTION

A vision–language model asked “what material is the glove?” about a 2250×1500 photograph must first encode the image into visual tokens. At any finite budget the glove may occupy a fraction of a token, and the evidence is gone before the language model runs. The model still *localises* it — its attention concentrates on the right region — but it cannot *perceive* it.

The standard remedy is attention-guided cropping: run the model once, read its attention over image positions, crop where the attention peaks, and re-encode that crop at higher density (Zhang et al., 2023; Shen et al., 2024; Anonymous, 2025). Every method in this family faces one decision that is rarely examined: **at which layer should the attention map be read?** Published methods pick a single mid-network layer or average a fixed block of late layers.

We show three things.

(i) The depth choice dominates. Cropping at the peak of one hand-picked mid layer scores 24.6% and 33.9% on two checkpoints — the four-option chance rate, and far below not cropping at all (62.3%, 57.4%). The same pipeline with a better depth read-out scores 77.2% and 71.3%.

(ii) Everyone reads at the wrong depth, and we can say why. Masking text tokens’ attention to the image, layer by layer, shows that image-to-text transport is *complete* by layer 16 of 28: blocking

every layer from 16 onward flips no answers. Yet every method in this family — including ours — reads its map at layers 17–22. The attention these methods consume is a residue of where the model looked, not the channel carrying the evidence.

(iii) **Two allocation policies follow from that measurement**, with complementary strengths (§4, §5), and at native resolution they match a model spending $5.5\times$ the compute.

2 SETUP

Benchmarks. V*Bench (Wu & Xie, 2024) ($n = 191$; 115 single-instance, 76 cross-instance) and HR-Bench 4K/8K (Wang et al., 2024) ($n = 800$ each; 400 per stratum). Both are public; we build no datasets. Questions are four-way multiple choice, scored by the arg-max over option-letter log-probabilities, so results are deterministic.

The equal-compute bar. A two-pass method that localises at 300 tokens and answers on a crop at 300 tokens spends 600 visual tokens. We therefore compare against `uniform@600`: the same budget spent on one higher-resolution pass. Published work in this area does not budget-match; §6 shows what that hides. Where pruning changes cost mid-network we measure **token-layers** (tokens $\times$ layers traversed); the bar is $600 \times 28 = 16,800$.

Stratification. Every result is reported separately for *single-instance* questions (one object’s property) and *cross-instance* questions (a relation between two). §6.2 shows these behave oppositely and that pooling them conceals it.

Statistics. Paired item bootstrap, 8000 resamples. Learned components use GroupKFold(5) by item $\times$ 3 seeds. A claim must hold on at least two checkpoints or it is reported as rejected.

3 WHERE THE IMAGE REACHES THE ANSWER

We install a hook that masks a chosen set of query positions from attending to the image-token columns, at chosen layers only, and measure the KL divergence at the output.

Table 1: Causal masking of text-to-image attention (Qwen3-VL-2B / Qwen2-VL-7B, $n = 40$).

mask scope (all layers)	KL at output	answers flipped
the answer token ’s row only	0.025	3%
all text rows after the image	1.931	42% / 45%

Blocking the answer-emission token from seeing any image token, at every layer, changes almost nothing (Table 1). Visual evidence reaches the answer through the *text* tokens, which consolidated it upstream. Masking one layer at a time for all text rows localises that consolidation: it peaks at L11 (Qwen3) and L14–15 (Qwen2), prefix masks $L_0..\ell$ saturate by $\ell = 12$ –16, and suffix masks $L_\ell..end$ fall to ≈ 0 from $\ell = 16$ (0.014 and 0.029; zero answer flips from $\ell = 20$). On a 36-layer checkpoint the same behaviour appears at L21, i.e. the boundary transfers as $0.57\times$ depth.

The consequence. The read-out band used by every method in this family (L17–20 on Qwen3, L19–22 on Qwen2) lies *entirely after* the transport window. Those layers’ image attention is causally inert ($KL \approx 0.001$). This predicts, and we confirm, three things: depth *re-weighting* helps while depth *selection* never does (§4); tokens can be dropped at the boundary for free (§5); and placement rules that all read the same inert depth separate only modestly.

Selection stops mattering exactly at the boundary. Before it, the choice of which tokens to keep is worth +8.4 and +8.9 points over random. At and after it, ranking is worth nothing: random keep equals attention keep on both models at every keep ratio. The boundary is not merely where pruning becomes cheap — it is where selection ceases to matter.

4 POLICY I: DEPTH RE-RANKING

Localise at 300 tokens with the full prompt (the read-out must be taken at the answer-emission position; without the answer instruction the map loses 36–42 points). Read the final prompt token’s attention over image cells at all 28 layers. Score each cell by a single closed-form ridge solve,

$$s(\text{cell}) = \sum_{\ell} w_{\ell} \log A_{\ell} + \sum_{\ell} v_{\ell} \text{rank}_{\ell} + a \log \text{nb} + \text{geometry},$$

fit on as few as 50 boxed examples. Crop $W=0.25$ at the arg-max, re-encode at 300 tokens, answer.

The fitted w_{ℓ} is the depth filter. It takes both signs: it assigns *negative* weight to 7 of the 11 layers the conventional block-mean read-out adds at +1, and its largest positive weight falls in the band where attention first becomes question-conditioned. There is no boosting, no learning rate, no early stopping.

5 POLICY II: TRANSPORT-SCHEDULED RESOLUTION

Because visual tokens are inert past the boundary, discard them there and spend the saving on pixels: encode at 900 tokens, run full width through L0–L16, keep the top 10% at L16, run L17–27 narrow. Token-layers $900 \times 17 + 90 \times 11 = 16,290$ — 96% of the bar, for $1.5 \times$ the resolution. No crop.

Pruning is free: 0.0–0.5 points in 9 of 10 measured cells, at every keep ratio (10/25/50%), with any ranking including random. Output probabilities change on 99% of items while the arg-max changes on 2% — the tokens are inert, not absent.

An identity. Across 4 checkpoints $\times$ 3 benchmarks, TSR’s gain equals the checkpoint’s *measurable resolution headroom* $\text{acc}(900) - \text{acc}(600)$ on that stratum: $r = 0.966$, slope 1.01, mean absolute error 0.7 points over 10 stratum-cells. The headroom needs two baseline runs and no method, so TSR’s benefit is predictable before deployment.

6 RESULTS

Table 2: All placement rules in one run, V*Bench, every arm at ~ 600 visual tokens. $\checkmark = 95\%$ CI clear.

method	Qwen3-VL-2B			Qwen2-VL-7B		
	single	cross	ALL	single	cross	ALL
uniform@600 (bar)	62.3	65.8	63.7	57.4	59.2	58.1
fixed layer L14	24.6	57.9	37.9	33.9	44.7	38.2
block-mean arg-max (ViCrop-style)	64.0	59.2	62.1	67.0	47.4	59.2
LASER (Anonymous, 2026)	64.0	65.8	64.7	65.2	55.3	61.3
gate-max (ours, no training)	72.8	63.2	68.9	63.5	56.6	60.7
ridge depth filter (ours)	77.2	65.8	72.6	71.3	68.4	70.2
<i>oracle crop (upper bound)</i>	96.5	75.0	87.9	93.9	77.6	87.4

The ridge filter beats every published rule on every cell where that rule was run — 9 of 9 paired contrasts, all CIs clear: +3.8 to +15.9 against ViCrop-style, +7.9 and +8.9 against LASER, +31.9 and +34.6 against fixed-layer. At matched compute the published rules are worth -1.6 to $+3.1$ against simply spending the tokens on resolution.

6.1 NATIVE DYNAMIC RESOLUTION

The field publishes these benchmarks at the model’s native dynamic resolution. On V*Bench that is a median of 3,290 visual tokens — $5.5 \times$ our bar — and 76.3% accuracy.

Three readings. Our methods are *statistically indistinguishable* from native resolution at 18% of its compute, while the published rules are *significantly worse* than native. Per stratum, the ridge filter at 600 tokens matches native on single-instance (+2.6) and TSR matches it on cross-instance (−2.6).

Table 3: Against native dynamic resolution, Qwen3-VL-2B, V*Bench, $n = 190$. Our arms use 18% of native’s compute.

arm	visual tokens	single	cross	vs. NATIVE (pooled)
NATIVE dynamic resolution	3,290	74.6	78.9	—
uniform@600 (bar)	600	62.3	65.8	−12.6 [−19.5, −5.8] ×
block-mean arg-max	600	64.0	59.2	−14.2 [−22.6, −5.8] ×
LASER	600	64.0	65.8	−11.6 [−20.0, −3.2] ×
TSR (ours)	600	65.8	76.3	−6.3 [−12.6, +0.0]
ridge filter (ours)	600	77.2	65.8	−3.7 [−12.1, +4.2]
<i>oracle crop</i>	600	96.5	75.0	+11.6 [+4.2, +18.9] ✓

And a perfectly placed 600-token crop *beats* native by +11.6 pooled and +21.7 on single-instance: placement is worth more than 5.5× the compute.

6.2 A SCOPE LAW

Across 14 cells (3 benchmarks × 4 checkpoints × 2 strata), attention-guided cropping helps single-instance perception in 7/7 and is non-positive on cross-instance in 6/7, significantly negative in three (−6.0, −8.8, −8.8). This holds for *every* method in the family, ours included: a tight crop magnifies one object and discards the other. TSR shows the opposite profile — positive on both strata in 6/7 cells with no significant loss anywhere — but gives up 3–10 points of single-instance gain.

The ceiling explains why cropping cannot be repaired here. With *perfect* placement, cross-instance still fails 25% and 22% of the time against 3.5% and 6.1% for single-instance. Cross-instance error is not predominantly perceptual, and no crop, multi-crop, composition or pruning schedule reached it (§7).

7 WHAT DOES NOT WORK

Eleven designs failed to make cropping serve cross-instance questions, each for a diagnosed reason: per-item window sizing; peak-spanning and mass-containment windows; joint (cell, window) re-ranking under a coverage objective (a metric artefact — coverage is monotone in window size); expected-accuracy window selection; a downscaled scene composed with the crop, which dilutes it by 5–9 points *even at oracle placement*; multi-crop at matched budget, where the second crop genuinely converts (+7.8 ✓) but halving each crop costs −11.3; and the same at full resolution funded by pruning. Depth *selection* rules ported from the LLM literature — DoLa, SLED, confident-layer decoding, attention-JSD, rank-stability — all fail, for one reason: in these models final-layer entropy rises on 54–88% of items while per-layer accuracy still climbs monotonically to the last layer. There is no better layer to select.

8 LIMITATIONS

$n = 191$ on V*Bench gives per-stratum CI half-widths near 10 points; no method is significantly positive on both strata in any single cell, which is a power limitation as much as a method one. The ridge filter needs ~50 boxed examples and two forward passes (1.4–2.0× wall-clock); TSR needs neither. Our two core checkpoints share a model family. The ridge filter’s pooled margin over TSR is concentrated on the two cells it was developed on. Pruning at the boundary is free in 9 of 10 cells; the exception (HR-Bench 8K, cross-instance, −1.8 ×) is real.

9 CONCLUSION

Attention-guided cropping reads the model’s attention at a depth where that attention no longer carries the image to the answer. Measuring where it does — a causal masking experiment, not a heuristic — yields both a better read-out and a free pruning point, and the two together match a

model running at $5.5\times$ the visual-token budget. The remaining gap is placement: perfect placement at 600 tokens exceeds native dynamic resolution by 11.6 points.

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